

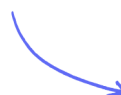
IGCSE • Cambridge (CIE) • Maths

 2 hours  25 questions

Target Test

Created 29/05/26

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Total Marks

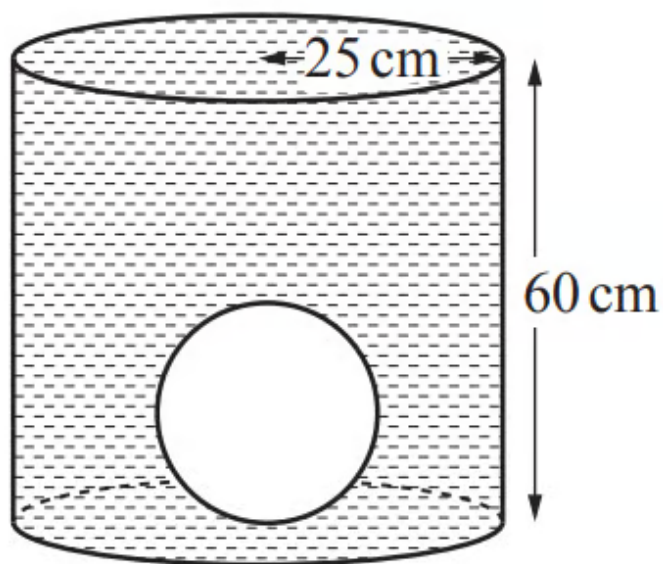
/121

- 1 (a)** Show that the volume of a metal sphere of radius 15 cm is $14\,140\text{ cm}^3$, correct to 4 significant figures.

[The volume, V , of a sphere with radius r is $V = \frac{4}{3} \pi r^3$.]

(2 marks)

- (b)** i) The sphere is placed inside an empty cylindrical tank of radius 25 cm and height 60 cm. The tank is filled with water.

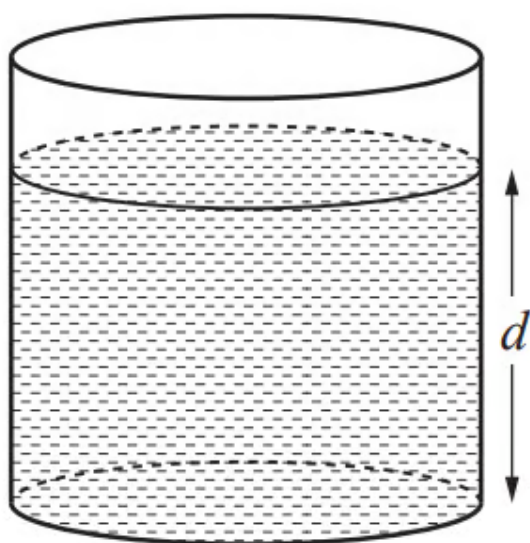


**NOT TO
SCALE**

Calculate the volume of water needed to fill the tank.

..... cm^3 [3]

ii) The sphere is removed from the tank.



NOT TO
SCALE

Calculate the depth, d , of water in the tank.

$d =$ cm [2]

(5 marks)

- 2 A circle centre O has radius 9 cm.

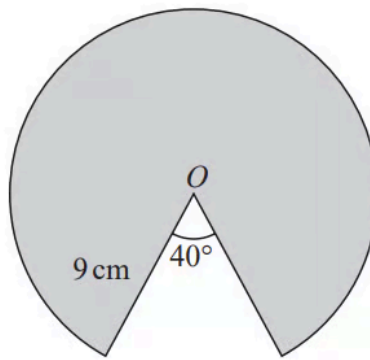


Diagram **NOT**
accurately drawn

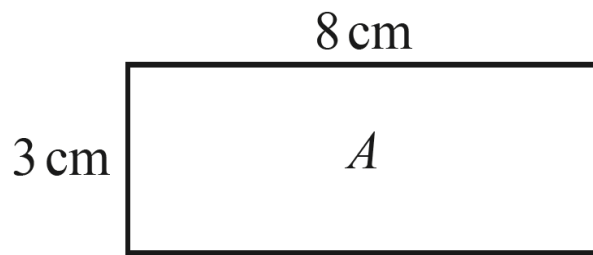
Calculate the perimeter of the shaded sector of the circle.

Give your answer as a simplified expression in terms of π .

..... cm

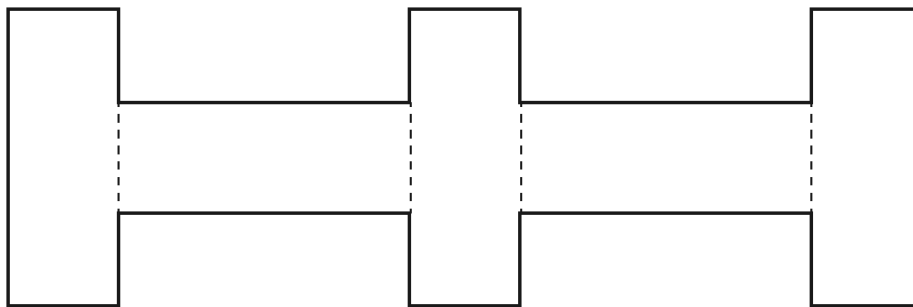
(4 marks)

- 3 Rectangle A measures 3 cm by 8 cm.



NOT TO
SCALE

Five rectangles congruent to A are joined to make a shape.



NOT TO
SCALE

Work out the perimeter of this shape.

..... cm

(2 marks)

- 4 (a) A cylinder with radius 6 cm and height h cm has the same volume as a sphere with radius 4.5 cm.

Find the value of h .

[The volume, V , of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.]

$h =$

(3 marks)

- (b) A solid metal cube of side 20 cm is melted down and made into 40 solid spheres, each of radius r cm.

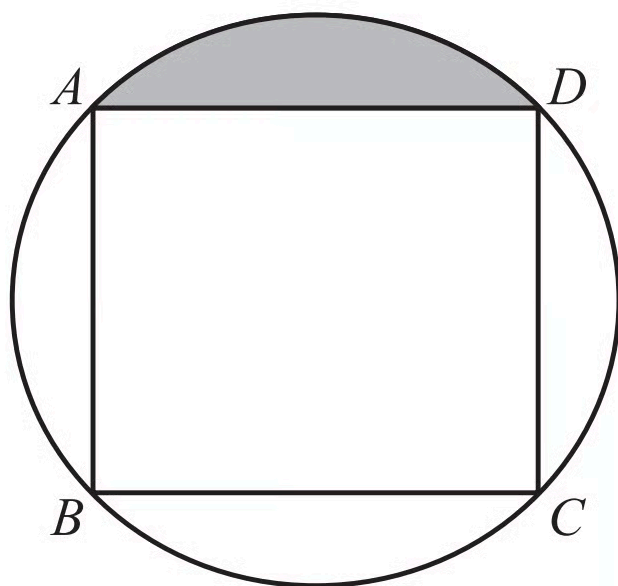
Find the value of r .

[The volume, V , of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.]

$r =$

(3 marks)

5 (a)



NOT TO
SCALE

The vertices of a square $ABCD$ lie on the circumference of a circle, radius 8 cm. Calculate the area of the square.

..... cm²

(2 marks)

- (b) i) Calculate the area of the shaded segment. Give your answer as a simplified expression in terms of π .

..... cm² [3]

- ii) Calculate the perimeter of the shaded segment. Give your answer as a simplified expression in terms of π .

..... cm [4]

(7 marks)

- 6 (a)** A solid metal cylinder has a volume of $450\pi \text{ cm}^3$.

The base of the cylinder has a radius of 5 cm.

Calculate the height of the cylinder.

..... cm

(2 marks)

- (b)** The solid metal cylinder is melted down and made into 2990 cubes, each with a side length of 0.75 cm.

Find the volume of metal **not** used in making these cubes.

..... cm^3

(2 marks)

- 7** A solid cylinder has radius x cm and height $\frac{7x}{2}$ cm. The surface area of a sphere with radius R cm is equal to the total surface area of the cylinder.

Find an expression for R in terms of x .

[The surface area, A , of a sphere with radius r is $A = 4\pi r^2$.]

$R =$

(3 marks)

- 8 Calculate the area of the sector of a circle with radius 65 mm and sector angle 42° .
Give your answer in square centimetres.

..... cm^2

(3 marks)

- 9 (a) A solid metal cone has radius 1.65 cm and slant height 4.70 cm.

Calculate the **total** surface area of the cone.

[The curved surface area, A , of a cone with radius r and slant height l is $A = \pi rl$.]

..... cm^2

(2 marks)

- (b) Calculate the volume of the cone.

[The volume, V , of a cone with radius r and height h is $V = \frac{1}{3} \pi r^2 h$.]

..... cm^3

(4 marks)

- (c) A metal sphere with radius 5 cm is melted down to make cones identical to this one.

Calculate the number of complete identical cones that are made.

[The volume, V , of a sphere with radius r is $V = \frac{4}{3} \pi r^3$.]

(4 marks)

- 10** A gold model is made.
This model is a prism with a cross-section of area 77.44 cm^2 .
- This gold model is 15 mm thick.
One cubic centimetre of gold has a mass of 19 grams.
- Calculate the mass of the gold model in kilograms.

..... kg

(3 marks)

- 11** The diagram shows a circle with centre O

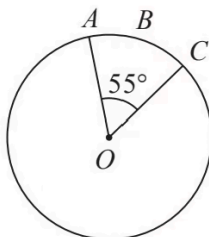
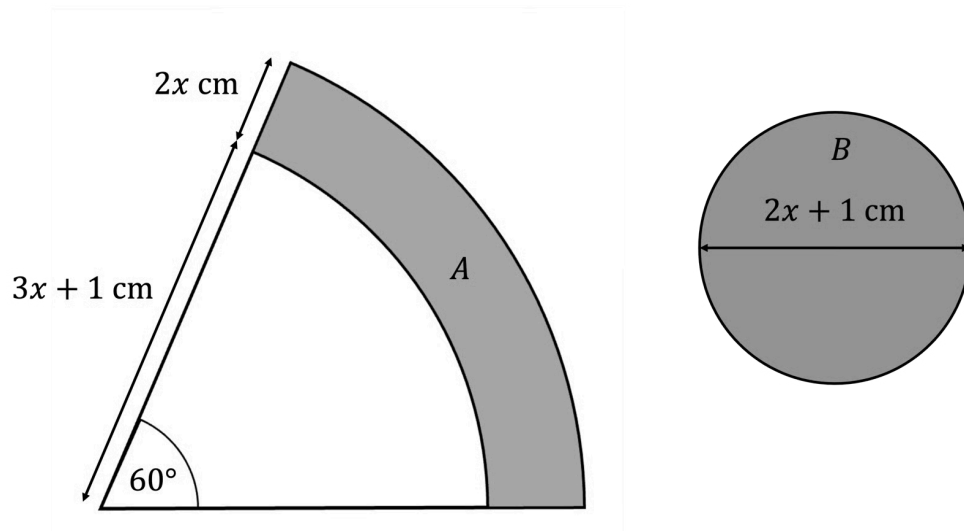


Diagram **NOT**
accurately drawn

- A , B and C are points on the circle so that the length of the arc ABC is 5 cm.
- Given that angle $AOC = 55^\circ$
- work out the area of the circle.
- Give your answer correct to one decimal place.

(4 marks)

12



The diagram above shows two shaded shapes, A and B .

Shape A is formed by removing a sector with radius $(3x + 1) \text{ cm}$ from a sector with radius $(5x + 1) \text{ cm}$.

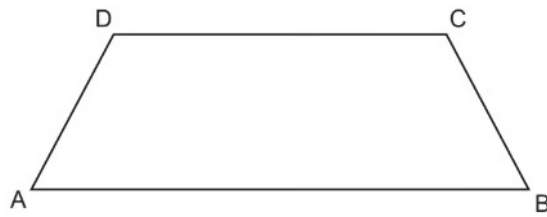
Shape B is a circle with diameter $(2x + 1) \text{ cm}$.

The two shaded sections have equal areas.

Find the value of x .

(6 marks)

13 ABCD is a trapezium.



Not to scale

The perimeter of the trapezium is 56 cm. The ratio $AD : AB : DC : BC = 5 : 12 : 6 : 5$.

Calculate the area of the trapezium.
Show your working.

.....cm²

(7 marks)

14 The total perimeter of a semicircle is 19.02 cm.

Calculate the radius of the semicircle.

..... cm

(3 marks)

15

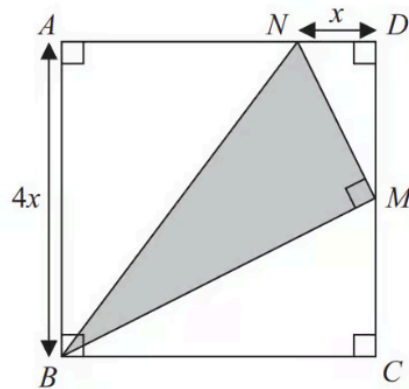


Diagram **NOT**
accurately drawn

$ABCD$ is a square with a side length of

M is the midpoint of DC .

N is the point on AD where $ND = x$

BMN is a right-angled triangle.

Find an expression, in terms of x , for the area of triangle BMN .

Give your expression in its simplest form.

(4 marks)

- 16 A solid metal sphere with radius 6 cm is melted down and all of the metal is used to make a solid cone with radius 8 cm and height h cm.

i) Show that $h = 13.5$.

[The volume, V , of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.]

[The volume, V , of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.]

[2]

ii) Calculate the slant height of the cone.

..... cm [2]

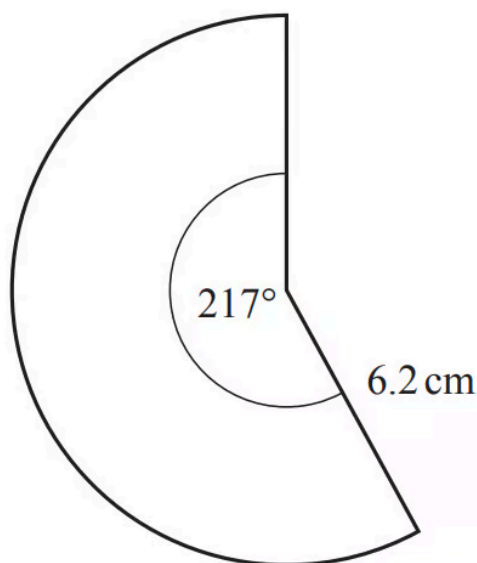
iii) Calculate the curved surface area of the cone.

[The curved surface area, A , of a cone with radius r and slant height l is $A = \pi rl$.]

..... cm² [1]

(5 marks)

17



NOT TO
SCALE

The diagram shows a sector of a circle with radius 6.2 cm and sector angle 217°.

Calculate the area of this sector.

..... cm²

(2 marks)

18 A pipe is full of water.

The cross-section of the pipe is a circle, radius 2.6 cm.

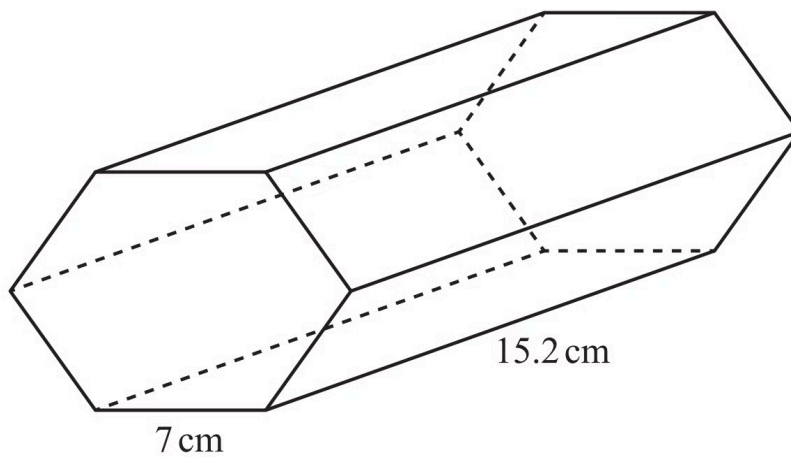
Water flows through the pipe into a tank at a speed of 12 centimetres per second.

Calculate the number of litres that flow into the tank in one hour.

..... litres

(3 marks)

19 (a)



NOT TO
SCALE

The diagram shows a solid prism with length 15.2 cm. The cross-section of this prism is a **regular** hexagon with side 7 cm.

Calculate the volume of the prism.

..... cm³

(5 marks)

(b) Calculate the total surface area of the prism.

..... cm^2

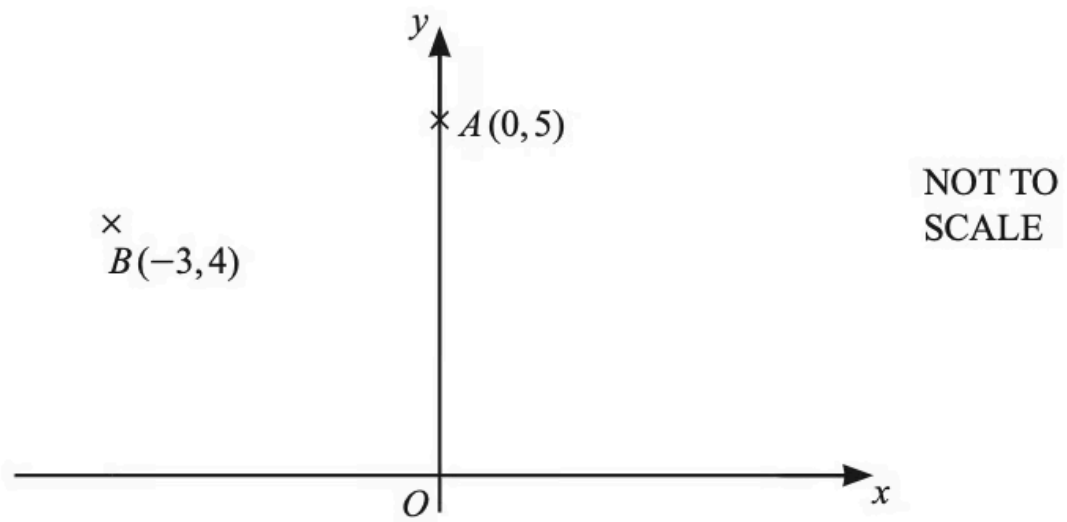
(3 marks)

20 Calculate the area of a circle with radius 5.1 cm.

..... cm^2

(2 marks)

21

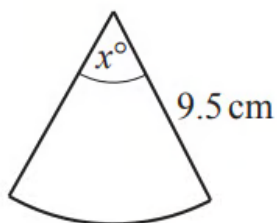
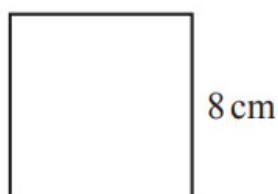


A and B lie on a circle, centre O .

Calculate the length of the arc AB .

(8 marks)

22



NOT TO
SCALE

The diagram shows a square with side length 8 cm and a sector of a circle with radius 9.5 cm and sector angle x° . The perimeter of the square is equal to the perimeter of the sector.

Calculate the value of x .

$x = \dots\dots\dots$

(3 marks)

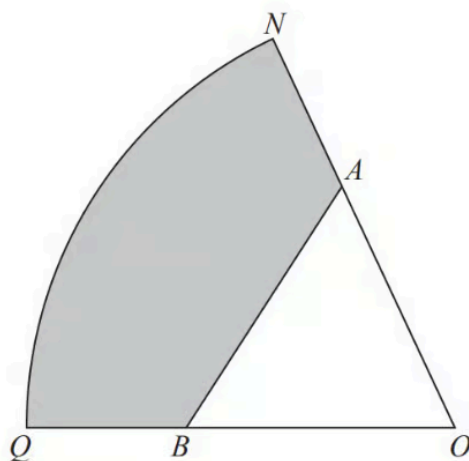
23 A pipe is completely full of water.

Water flows through the pipe at a speed of 1.2 m/s into a tank. The cross-section of the pipe has an area of 6 cm^2 .

Calculate the number of litres of water flowing into the tank in 1 hour.

$\dots\dots\dots$ litres

(4 marks)



ONQ is a sector of a circle with centre O and radius 11 cm.

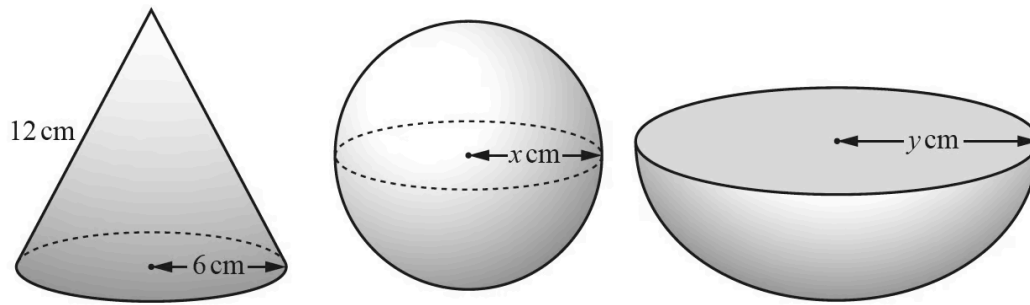
A is the point on ON and B is the point on OQ such that AOB is an equilateral triangle of side 7 cm.

Calculate the area of the shaded region as a percentage of the area of the sector ONQ .
Give your answer correct to 1 decimal place.

(5 marks)

25 (a)

NOT TO
SCALE



The diagram shows three solids.

The base radius of the cone is 6 cm and the slant height is 12 cm.

Show that the total surface area of the cone is $108\pi \text{ cm}^2$.

[The curved surface area, A , of a cone with radius r and slant height l is $A = \pi rl$.]

(2 marks)

(b) The radius of the sphere is x cm and the radius of the hemisphere is y cm.

The **total** surface area of each solid is the same.

Find the value of x and the value of y .

Give the value of x as a simplified surd.

[The surface area, A , of a sphere with radius r is $A = 4\pi r^2$.]

$x = \dots\dots\dots$

$y = \dots\dots\dots$

(4 marks)